DATA601 Report

Development of Intensity Prediction/Attenuation Equation for GeoNet's Felt RAPID Report (FRR) data

Benjamin Woods-66281658
Zheng Chao-21671773

Abstract

Intensity prediction equations are useful for conducting earthquake risk assessment and rapid assessment after an earthquake event. In New Zealand, Felt Rapid Report (FRR) data collected by GeoNet provides a large crowdsourced dataset for recording the intensity of the perceived shaking. This is a survey comprised of 6 cartoons which correspond to value of 3-8 on the MMI scale. This study aims to utilize the FRR data to determine an effective and physically meaningful method for simulating the relationship between MMI and the distance from the source, and to assess the influence of key confounding variables such as magnitude, depth, and site conditions (Vs30). We evaluated a series of regression and classification models, including linear models, tree-based methods and ensemble models. An 80/20 training/testing split was used with group aware splitting to avoid data leakage and particularly beneficial for large datasets, where 80% can still represent a vast amount of data for training (Kurnia, 2024). The model performance was evaluated using standard metrics including R^2, MSE, accuracy, and macro F1 value, as well as the domain-specific indicators of percentage of predicted values within ±1 and ±2 MMI units. Additionally, the intensity maps of the predictions were examined to ensure their physical rationality, particularly the monotonic attenuation of intensity with distance. Although some complex models achieved good numerical performance, many models failed to generate intensity maps that conformed to physical laws. Ridge regression with a second order polynomial feature transform was the model selected because it achieved the best balance among prediction accuracy, interpretability, and spatial consistency. The results indicated that when modeling noisy crowdsourced intensity data, relatively simple and well-constrained models can outperform more complex methods and emphasized the importance of combining quantitative metrics with physical verification in earthquake modeling.

Table of Contents

1. Introduction

GeoNet collects crowdsourced data on the intensity of ground shaking experienced during earthquakes across New Zealand. The primary system used to collect this data is the Felt Rapid Report (FRR) system which collects user inputs based on cartoon depictions of 6 levels of shaking intensity during an earthquake (GeoNet, n.d.). These levels are designed to match the Modified Mercalli Intensity (MMI) scale. The MMI scale categorises shaking intensity empirically based on the effects the shaking has on people, structures and land; there are 12 ordinal levels ranging from Not felt(I) to Extreme(XII). Shaking intensity is spatially variable and is different at each point on the surface (USGS, 2021). Table 1 summarizes the Modified Mercalli Intensity Scale according to GeoNet.

Table 1. MMI scale from GeoNet

MMI	Intensity	Description
1	unnoticeable	Barely sensed only by a very few people.
2	unnoticeable	Felt only by a few people at rest in houses or on upper floors.
3	weak	Felt indoors as a light vibration. Hanging objects may swing slightly.
4	light	Generally noticed indoors, but not outside, as a moderate vibration or jolt. Light sleepers may be awakened. Walls may creak, and glassware, crockery, doors or windows rattle.
5	moderate	Generally felt outside and by almost everyone indoors. Most sleepers are awakened and a few people alarmed. Small objects are shifted or overturned, and pictures knock against the wall. Some glassware and crockery may break, and loosely secured doors may swing open and shut.
6	strong	Felt by all. People and animals are alarmed, and many run outside. Walking steadily is difficult. Furniture and appliances may move on smooth surfaces, and objects fall from walls and shelves. Glassware and crockery break. Slight non-structural damage to buildings may occur.
7	severe	General alarm. People experience difficulty standing. Furniture and appliances are shifted. Substantial damage to fragile or unsecured objects. A few weak buildings are damaged.
8	extreme	Alarm may approach panic. A few buildings are damaged and some weak buildings are destroyed.
9	extreme	Some buildings are damaged and many weak buildings are destroyed.
10	extreme	Many buildings are damaged and most weak buildings are destroyed.
11	extreme	Most buildings are damaged and many buildings are destroyed.

Magnitude is a measurement of the size of an earthquake and there is a single value for an entire earthquake which measures the total energy released. Magnitude is a logarithmic scale with each whole number being a 10 times increase in seismic wave amplitude and a 32 times increase in total energy released. Higher magnitude earthquakes generally have higher intensities, but other factors influence how strong the shaking is such as depth, distance from hypocenter to observer, and local ground conditions (USGS, nd).

Being able to model how ground shaking intensity varies based on distance from an earthquake is important for fast assessment of where damage is likely after an earthquake, seismic risk analysis and evaluation of historical earthquakes. New Zealand has a large database of FRR data which can be easily accessed through the GeoNet API and a unique tectonic setting which provides the opportunity to develop a New Zealand specific intensity prediction model using modern data science methods.

The primary goal of this research project is to develop an intensity attenuation relation model for New Zealand based on FRR data, and will answer the following:

- What is the most effective way to model the relationship between MMI values and the distance from the earthquake hypocenter?
- Which confounding variables associated with earthquakes and their perceived intensity have a significant influence on this relationship?

Intensity prediction equations (IPE) are equations which take input variables such as distance, magnitude and depth and calculates a corresponding intensity. IPEs are useful as predicting intensity from ground motion requires the development of ground motion prediction equations (GMPE) and ground motion to intensity conversion equations (GMICE), which when used results in the combining of their uncertainties and the overall prediction being losing accuracy. An example of one of these equations is that of Dowrick and Rhoades (Dowrick, 2005):

Dowrick and Rhoades (2005): New Zealand – Main Region:

$I_{MMI} = 4.40 + 1.26M_W - 3.67\log(r^3 + d^3)^{1/3} + 0.012h + 0.409$

Where:

d = 11.78

r = distance to rupture

M_W = moment magnitude

h = earthquake focal depth

This equation takes the form of a mathematical equation with inputs of moment magnitude, depth and distance to rupture. These equations have the following differing measurements of distance that offer increasing levels of accuracy but increasing complexity of calculation:

1. Epicentral distance - The horizontal distance between the observer and the point on the Earth's surface directly above the earthquake hypocenter.
2. Hypocentral distance - The straight-line (3-D) distance between the observer and the earthquake hypocenter, accounting for focal depth.
3. Closest distance to Rupture - The shortest distance from the observer to any point on the fault rupture surface.

In this report hypocentral distance will be used as the GeoNet API does not have information about the rupture plane of earthquakes. Notably on the Dowrick and Rhoades equation and most other IPEs there is no term for local site effects, according to Borchedt, there is a large intensity amplification effect from soft near surface sediments (Borcherdt, 1970). This will be addressed using a vs30 measurement derived from a map of New Zealand which measures the speed of earthquake s waves through the ground 30 meters deep. This is captures how soft the ground is and acts as an easy accessible proxy to local conditions.

FRR data is noisy and subjective, this is because submissions are open to anyone and people's opinions differ on how strong an earthquake feels. These reports are geohashed into a grid of small squares which compile all reports from that square into a single dictionary, this is done by GeoNet for privacy reasons. This helps deal with the noise but a decision has to be made on how to average these reports to a single value. The reports are also not uniform and are clustered in cities and towns where people are present, this means that there is a lot of areas that are not covered by reports and this has to be factored into the modeling pipeline.

2. Data

The dataset consists of three components: Felt Rapid Report (FRR) data, earthquake statistic data, and Vs30 data. The FRR and earthquake statistic data are provided for each earthquake recorded by GeoNet, and the dataset required careful selection of what earthquakes to include.

2.1 Felt Rapid Reports

The primary data source for this research project is GeoNet's Felt Rapid Report (FRR) data. This data is crowdsourced through an online survey where users select from a set of 6 cartoon images depicting increasing levels of shaking. An example of one of these cartoons can be seen in Figure 1, this cartoon corresponds to MMI level 4. These cartoon images correspond with levels 3 to 8+ on the MMI scale, MMI levels 3 to 8 were chosen by GeoNet as below MMI 3 it is hard for people to feel an earthquake and above 8 requires an engineer's assessment. These FRR can be submitted at any time and GeoNet will assign them to specific earthquake events, the FRR are geohashed into a 1km-by-1km grid to provide privacy, this also reduces the noise in the dataset. This FRR data is provided for each individual earthquake event and is accessed through the GeoNet API, each earthquake felt file is provided in the GeoJSON format with each row being a cell in the 1km-by-1km grid and all the felt reports in that row being in a dictionary format. Each earthquake event is assigned a PublicID which allows lookup within GeoNets API and can be used to join these felt reports with other earthquake statistics.

Figure 1. FRR cartoon depicting MMI 4

2.2 Earthquake Statistics

A secondary data source for this research project was the earthquake statistics provided by GeoNet. This data is generated by GeoNet for each earthquake event and contains scientific measurements of earthquake features including location, magnitude, time, and depth. These features provide the necessary data to calculate the hypocentral distance, and provide the depth and magnitude which are typically used in Intensity Prediction Equations. Each earthquake event has a separate file and the corresponding PublicID is used to access this and merge with FRR data.

2.3 Vs30 Data

A supporting data source for this research project is Vs30 data for New Zealand. Vs30 is the shear-wave time averaged velocity to a depth of 30 meters. This represents the stiffness of the ground at a given location and is used as a proxy for local site effects as softer ground tends to amplify seismic waves which results in a stronger shaking and thus intensity, whereas stiffer ground reduces this effect (Boore, 1997). A raster file of Vs30 values over New Zealand was obtained from QuakeCoreSoft, this can then be sampled for values in the same geohashed grid used in the FRR data.

2.4 Earthquake Selection

The total number of earthquakes recorded in New Zealand is mostly comprised of weak earthquakes with 98.5 percent of earthquakes in the last 365 days being magnitude 4.0 or below according to GeoNet through quakesearch. These small earthquakes generate very low amounts of felt reports with a large amount of mmi 3 FRR entries. This means that several criteria must be applied to make sure that there is a large amount of felt reports and a wide range of FRR values. It is also important to gather earthquakes from an even distribution of tectonic regions within New Zealand to provide a variety in the data and reduce overfitting to the specific regions that have more earthquakes.

The temporal range was selected to be from 1 October 2016 to 15 December 2025, this is because the FRR system was released in September 2016 and the data was gathered in mid December 2025. The regions selected for inclusion were Gisborne, Hawkes Bay, Wellington and Canterbury. These regions were chosen as they correspond with the tectonic regions Hikurangi Subduction Zone, Wellington Fault, Kaikoura and parts of the Alpine Fault. The Hikurangi Subduction Zone produces many earthquakes and is capable of producing earthquakes above 8.0 (ECL, 2023), the Wellington and Alpine faults are cable of producing earthquakes up to 8.2 (Howarth et al., 2018), and the Kaikoura area is important to include because of the major 2016 7.8 magnitude earthquake and related aftershocks which provide large earthquakes for modeling purposes. A lower magnitude bound of 4.0 was chosen as this has a higher chance of having more felt reports and avoids the problem of weaker earthquakes only having mmi 3 felt reports.

A list of earthquakes that matched this criteria was then generated by using the GeoNet quake search tool and applying the constraints as filters. These earthquakes were then further filtered in python and were found to be an imbalanced data set of 262 earthquakes with there being more weaker than stronger ones. A balanced dataset was preferred so this list was then filtered to have 24 earthquakes maximum from each 0.5 magnitude bin between 4.0 and 6.0 with all above 6.0 being kept, this was done randomly with a reproducible seed. The remaining dataset consisted of 103 earthquakes.

Table 2. Earthquake Stats Variables:

Variable	Description	Type	Units	Source
PublicID	Primary key for each earthquake event	Categorical	text	GeoNet API
Magnitude	Earthquake magnitude	Continuous	Mw	GeoNet API
Depth	Earthquake depth	Continuous	Km	GeoNet API

Epicenter Location	Tuple of latitude and longitude of epicenter	Continuous	Degrees	GeoNet API

Table 3. FRR variables:

Variable	Description	Type	Units	Source
Report Location	Tuple of latitude and longitude of geohashed cell	Tuple of continuous variables	Degrees	GeoNet API
mmi	Max mmi in cell	Categorical	3-8	GeoNet API
count	Number of FRR in cell	Integer	-	GeoNet API
count_mmi	Dictionary of FRR with mmi as key and count as value	Dictionary	-	GeoNet API
Vs30	Shear-wave velocity at 30 m depth	Continuous	m/s	QuakeCoreSoft

2.5 Cleaning

The FRR data contained inconsistencies, variability and invalid data inherent in crowdsourcing and the following steps were applied before modeling. Records with missing values were removed to maintain a complete dataset. Records with mmi values outside the range of 3 to 8 were deleted as these are not possible to input in the FRR system and must be an error. All record geometries were checked to ensure they fall within the geographic bounds of New Zealand, defined as latitudes between −48° and −34° and longitudes between 165° and 180°. Records outside this range were considered invalid and removed. The FRR table and Stats table were merged by adding the PublicID into each FRR record using the file name, then using a join between them on the PublicID. This allowed for the epicentral distance to be calculated between the report location and the epicenter location using the Haversine formula, this epicentral distance then was used with the depth to calculate the hypo central distance using Pythagorean formula. Vs30 values were sampled from the QuakeCoreSoft Vs30 raster file and added to each record. The FRR table and the earthquake stats table were joined on PublicID, with the PublicID for FRR being provided through the filename.

A minimum number of reports per cell had to be selected to produce stable results and reduce noise. Selecting a minimum number of reports is a trade off between spatial coverage of cell intensity data and stability of cell mmi values, as the number of minimum reports increases the spatial coverage rapidly decreases as can be seen in Figure 2. It is important to have a large spatial coverage so the model does not overfit to just the main population centers. In Figure 3 it can be seen that the value and the spread of the standard deviation of cells drops quickly from 1 report and stabilizes at 5 to 10 reports. A minimum report count per cell was set at 5 to balance these factors.

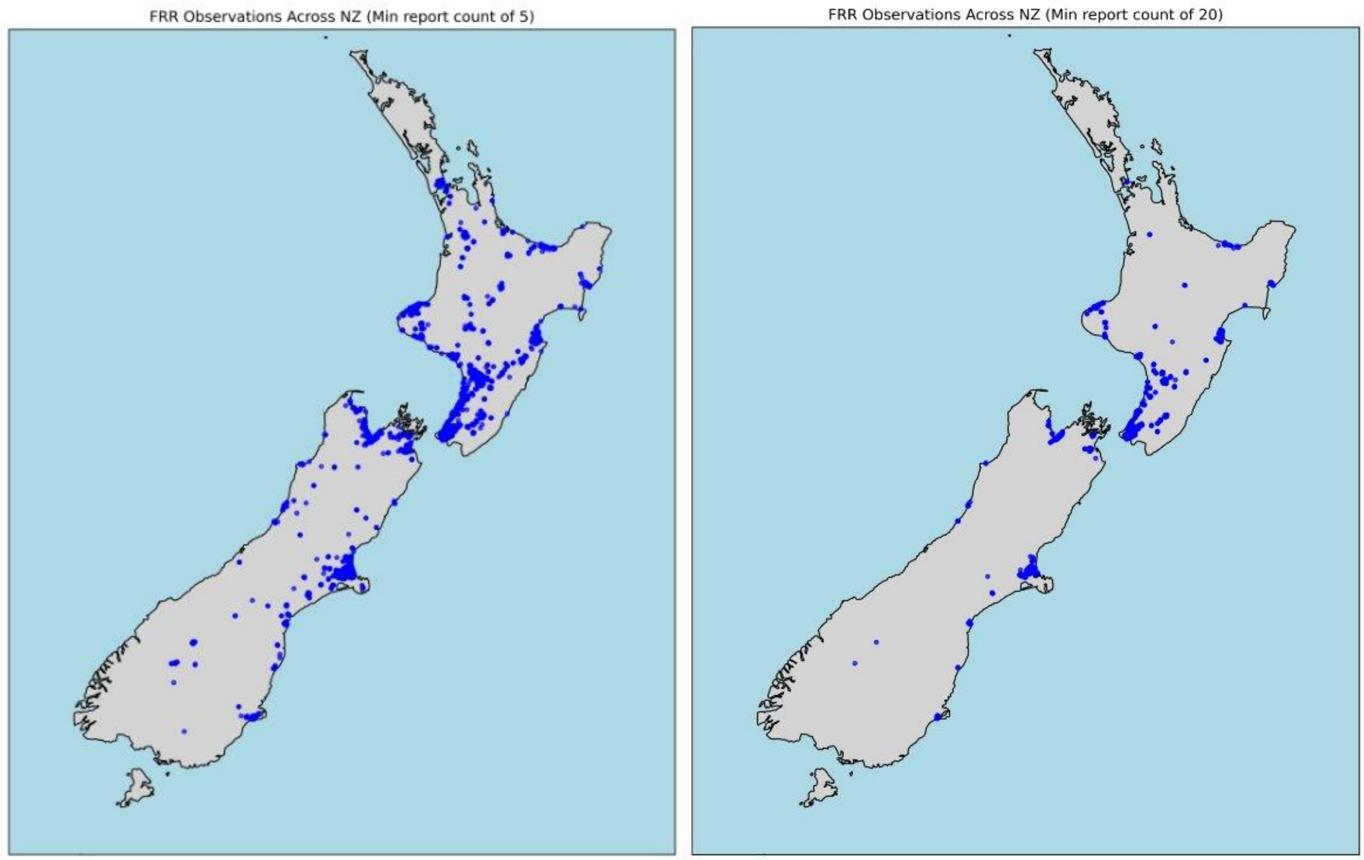

Figure 2. Decreasing spatial coverage with increasing min count

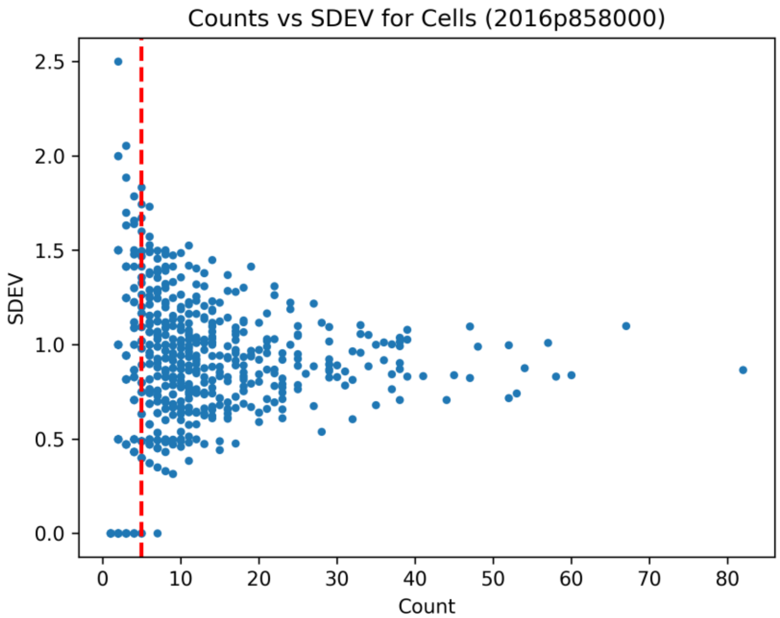

Figure 3. Counts vs Standard deviation for specific earthquake

Due to the aggregation of FRR into a geohashed grid of cells each cell can have multiple reports, this means that a measure of central tendency must be chosen to represent these cells. In Figure 4, a strong positive skew is observed for grid cells with mean MMI between 3 and 4, before changing to negative skew beyond around mean MMI 4. For mean MMI values above 5, the distributions are generally close to symmetric but exhibit a slight negative skew. The strong positive skew at low mean mmi values is from the lower bound of the reports being 3, this means that close to mmi 3 is the average value but it is being skewed by higher values but there are no possible lower values. The skew can then be seen flipping negative as this is reversed as after mean mmi 4 most reports are either mmi 3 or 4. This effect creates lines that go diagonally from positive to negative skew, this is caused by the mmi inputs from the FRR's being discrete values. The mmi scale is an ordinal scale and therefore the distances between levels are not known, for example the gap between mmi 3 and mmi 4 is not necessarily the same size as the gap between mmi 7 and mmi 8. A median measure of central tendency can produce cell mmi values of .5 differences and therefore produces invalid mmi values. For these reasons, the mode was chosen as the measure of central tendency. This was calculated per record in the FRR data and added as the variable mode.

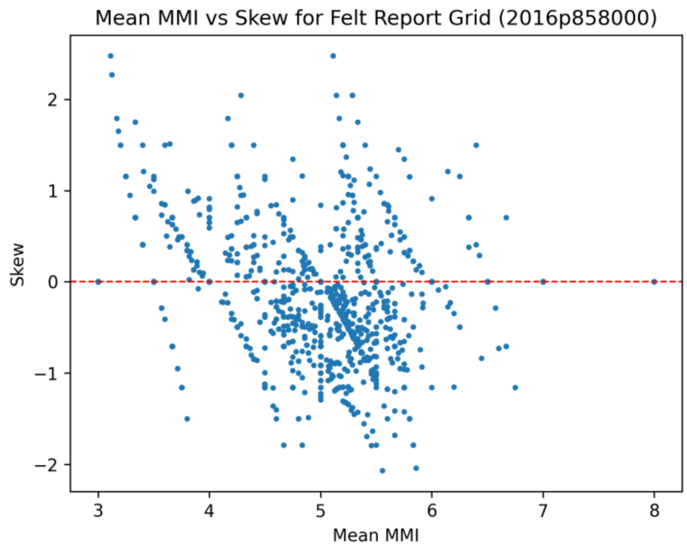

Figure 4. Mean MMI vs skew for specific earthquake

Table 4. Final FRR variables:

Variable	**Description**	**Type**	**Units**	**Source**
PublicID	Primary key for each earthquake event	Categorical	text	GeoNet API
Report Location	Tuple of latitude and longitude of geohashed cell	Tuple of continuous variables	Degrees	GeoNet API
mmi	Max mmi in cell	Categorical	3-8	GeoNet API
count	Number of FRR in cell	Integer	-	GeoNet API
count_mmi	Dictionary of FRR with mmi as key and count as value	Dictionary	-	GeoNet API

Vs30	Shear-wave velocity at 30 m depth	Continuous	m/s	QuakeCoreSoft
Magnitude	Earthquake magnitude	Continuous	Mw	GeoNet API
Depth	Earthquake depth	Continuous	Km	GeoNet API
Epicenter Location	Tuple of latitude and longitude of epicenter	Continuous	Degrees	GeoNet API
Epicentral Distance	Distance along the surface from the epicenter to report location	Continuous	Km	Calculated
Hypocentral Distance	Straight line distance from the hypocenter to report location	Continuous	Km	Calculated
Mode	Mode of the mmi counts	Categorical	3-8	Calculated
Region	Region the earthquake epicenter is located in.	Categorical	-	Geonet

3. Method

The methodology of this report.

3.1 Test Train Split

The modelling workflow of this research project followed a pipeline of data gathering, data cleaning, feature engineering, and then combining all of the individual earthquake FRR tables into a single large one. This table had a column PublicID which is the unique identifier of each earthquake event. A list of earthquake PublicIDs was kept, this was then randomly split into an 80/20 split, and the PublicIDs were used to split the larger table, this process ensured that the training and testing sets consisted of whole earthquake felt reports and there was no data leakage between.

3.2 Regression vs Classification

The relationship between local earthquake intensity and the explanatory variables can be approached as either a categorisation problem or a regression problem. Both approaches were used in this project, classifiers make sense to use as MMI is an ordinal variable and classifiers directly respect these categories. However, classifiers do not necessarily respect the ordered nature of this data, and misclassifications between far levels and close levels will reflect the same performance metrics of accuracy, precision and recall. Regression models were also used as earthquake shaking is a continuous physical process, this was achieved by treating MMI as an approximate continuous variable and then rounding to the nearest MMI level after predicting values. Regression models also allow for a clean and easy to interpret shaking map to be developed from the continuous data before rounding. As both classification and regression seem to have similar advantages and limitations many models from both groups were tried.

3.3 Class Imbalance

The earthquake magnitude classes are balanced in this dataset however the FRR MMI values are not, they are primarily MMI 3 or 4 with a very low proportion of MMI 6 or greater, this can be seen in Figure 5. This bias can make the model predict far more lower values or even just predict around the average value between 3 and 4. To combat this downsampling was applied to the larger categories to reduce their influence and balance the dataset. It aims to correct imbalanced data and thereby improve model performance (Murel, n.d.). this does result in losing some data but there is such a large amount available this should not be an issue. Another approach to balance the dataset was to apply the ‘class_weight = “balanced”’ in the options for the classifiers which had it. This increases the penalty for misclassifying underrepresented MMI categories, this allows the model to handle rare high-intensity observations without changing the underlying data distribution. These strategies are applicable to the MMI intensity prediction problem as rarer; higher intensity events are the most important to be able to predict and model correctly as they cause the most damage.

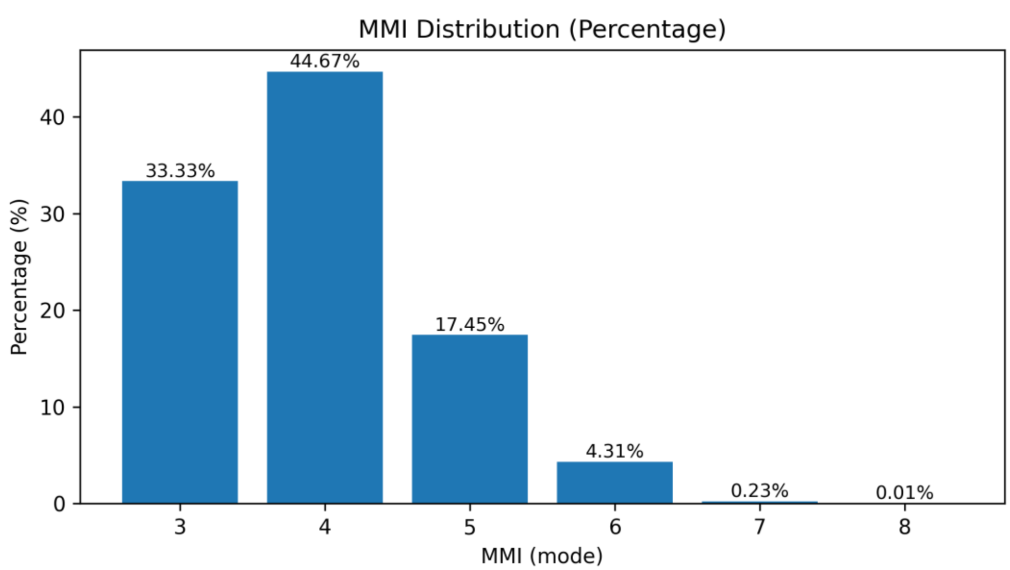

Figure 5. Distribution of FRR MMI values (percentage)

3.4 Modelling Approaches

13 different classification models and 26 different regression models were tested, the scikit-learn python package provided these. The relationship between shaking intensity (MMI) and predictor variables is inherently non-linear and is complex so will not be adequately modeled by a simple linear model. This led the model selection in the direction of models that can capture non linear relationships, however some linear models were attempted as a baseline and because if they were successful enough then they would be highly interpretable. Linear regression provides a simple and interpretable baseline that is suitable for modelling average intensity trends but struggles to capture the non-linear patterns of this problem. Decision trees are capable of modelling non linear relationships between distance and intensity, but they are sensitive to noise and tend to perform poorly for high MMI values, and can not predict outside of the categories present in the data. More flexible models, such as gradient-boosted methods and generalized additive models (GAMs), can capture non-linear relationships and have great usability while retaining some interpretability, although they carry a higher risk of overfitting (Masui, 2022).

3.6 Evaluation Strategy (/ Metrics)

Model performance was evaluated using macro precision, macro recall, macro accuracy, and macro F1 score for the classification models; and R^2 and MSE for regression models. Accuracy tells us the proportion of correct predictions made by the model out of all predictions. R^2 score represents the proportion of the variance in the dependent variable that is predictable from the independent variables (GeeksforGeeks, n.d.). For both modeling approaches additional evaluation metrics were calculated due to the practical domain requirements; these were percentages of predictions ±1 MMI unit and ±2 MMI units of the observed intensity value. These provide a way to compare classifiers against regressors and also overcome the issue discussed above about performance metrics not being able to capture the distance of misclassified observations.

4. Results:

39 modeling methods were tested by training them on 80 percent of the dataset and then testing on the other 20 percent, GroupShuffleSplit from scikit-learn was used to ensure that each earthquake was either in the testing or training group to avoid data leakage. The testing set was used to make MMI predictions using Vs30, depth, magnitude and hypo distance as predictors. These predictions were then used to calculate the metrics. Along with these metrics the model intensity predictions were spatially mapped onto a New Zealand map. For example, the HistGradientBoosting, the polynomial ridge. These maps were used as a sanity check for the model, checking if the predictions followed the basic physical constraint of intensity monotonically decreasing with distance. Many models had good performance metrics but having intensity predictions increase with distance, exhibit a wave like pattern, be full of noise, or be all very close to a single average value. These models were all disqualified if the mapping could not be fixed, no matter how high the metrics were.

An example of a model with high metrics but a bad map was HistGradientBoosting which had an R^2 of 0.37, MSE of 0.52 and 86.3 percent within 1 MMI level. These metrics look good but when mapped it predicts noise around MMI 3 for all close values, then noise around 4 for further values before dropping back down to 3. This violates the basic physical principle of earthquake shaking intensity decreasing with distance and is therefore not a good model.

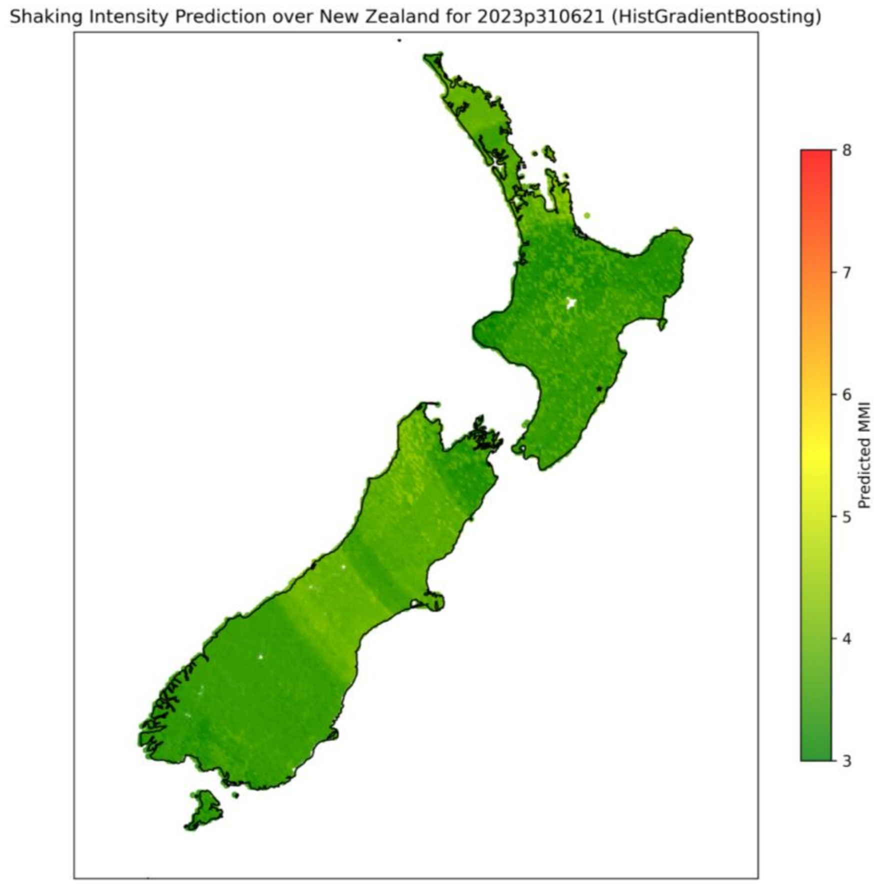

Figure 6. Predicted MMI shake map over New Zealand (HistGradientBoosting)

After checking the maps and removing regression models with negative R^2 values 4 regression based models and 4 classification based models were left. The performance metrics of these models can be seen in table 5 and table 6 below.

Table 5. Regression Models

Model	R^2	**MSE**	**% within 1 MMI**	**% within 2 MMI**
Polynomial degree 2 Bayesian Ridge	0.317	0.574	85.2	99.5
Polynomial degree 2 Elastic Net	0.268	0.615	85.3	99.5

Polynomial degree 2 Ridge	0.317	0.574	85.2	99.5
Monotonic HistGradientBoostingRegressor	0.439	0.384	90.6	99.1

Table 6. Classification Models

Model	**Accuracy**	**Macro F1**	**% within 1 MMI**	**% within 2 MMI**
Ordinal Logistic	0.467	0.188	39.2	92.4
Ordinal Ridge	0.485	0.189	46.9	92.8
Polynomial degree 2 Logistic Regression	0.480	0.200	37.3	92.7
Polynomial degree 2 Ordinal Ridge	0.500	0.184	47.3	92.8

From these statistics the two best performing regression models are Polynomial degree 2 Ridge and Monotonic HistGradientBoostingRegressor. Polynomial degree 2 Ridge consisted of a two step pipeline with a polynomial degree 2 transformation applied first, this expanded the features to include each multiplicative interaction term up to degree 2. For example, features such as Vs30 * depth, magnitude^2 etc. would be included. This left 14 features, after which ridge regression was applied normally. Monotonic HistGradientBoostingRegressor was a HistGradientBoostingRegressor with a monotonic decreasing constraint being placed on the hypocentral distance feature. None of the classification models have more than 50% within 1 MMI, because of this none were considered further, if discrete MMI categories are desired then regressors can be rounded. Polynomial degree 2 Ridge regression is the simpler model here, and it is highly interpretable and parsimonious so was chosen.

Polynomial Ridge Regression:

Figure 7 shows the intensity map produced by Polynomial degree 2 Ridge, it can clearly be seen that close to the earthquake there is a higher MMI of above 5 before decaying to 3 with distance, the effects of Vs30 can be seen showing the impact of local ground conditions on intensity. Even at extremely far distances this model does not decay below 3, this is due to the limitations of the FRR input data which has a minimum value of 3 while the MMI system goes down to 1; this is discussed further later in this report. The effects of vs30 can be seen as well, this is the reason for non symmetrical decay of MMI, however the model has learnt that higher vs30 causes higher MMI however most evidence points that higher vs30 should cause lower intensity. This is likely due to the nature of FRR data and cities in general being built on areas with softer soils and thus lower vs30. This leads to far more data points having a low vs30 causing an unbalance in the dataset.

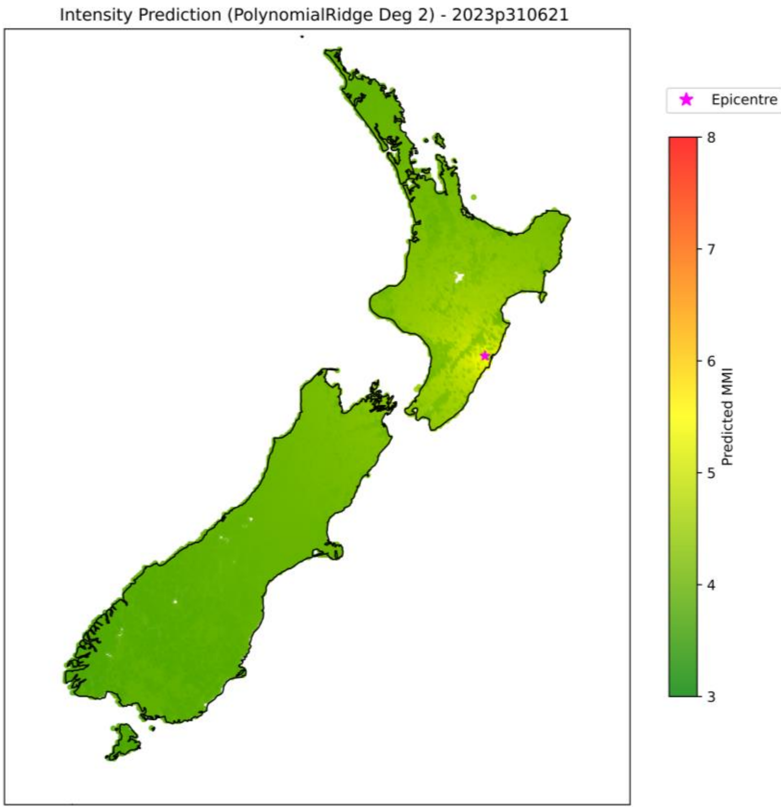

Figure 7. Predicted MMI shake map over New Zealand (Polynomial Ridge)

Figure 8 is the residual vs predicted MMI plot degree 2 Polynomial Ridge model, the first thing to notice about this is that the residuals are aligned in diagonal lines. This is because MMI values are categorical variables and mode was selected as the measure of central tendency for the FRR data. Outside of these lines the residuals are generally randomly scattered around 0, higher values seem to have more variance. There are an even amount of positive and negative residuals and show a slight downward slope with higher predictions of mmi.

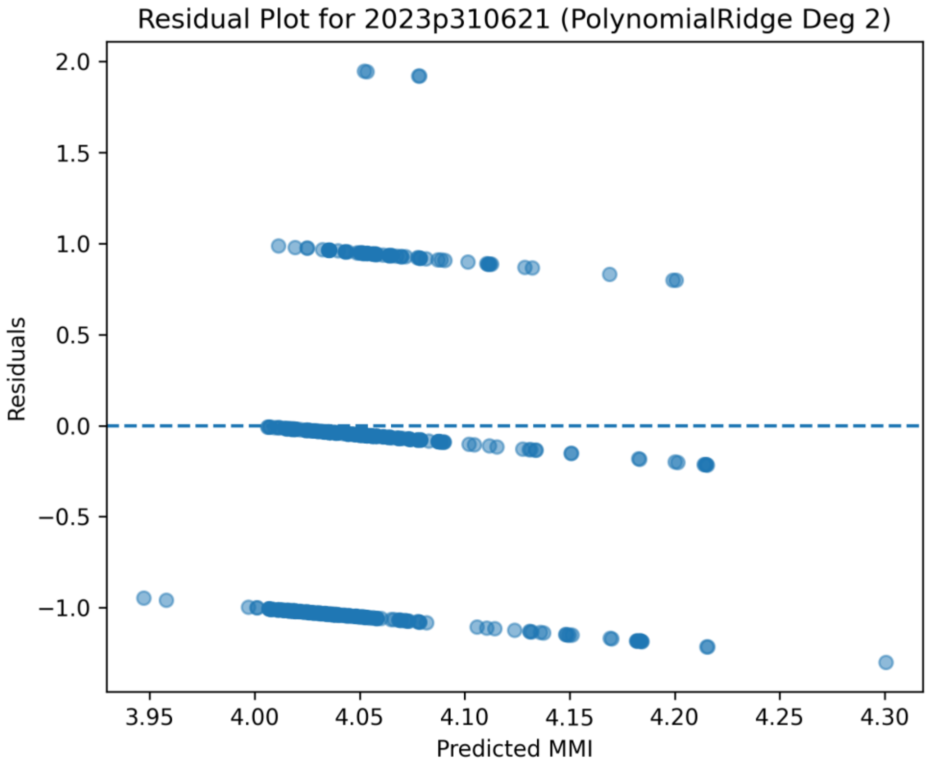

Figure 8. Residual plot for Polynomial Ridge

Monotonic HistGradientBoostingRegressor:

Figure 9 shows the intensity map produced by Monotonic HistGradientBoostingRegressor, the MMI around the north island is generally between 4 and 5 with a small amount of noise caused by vs30. The MMI then drops to below 4 and then reaches 3. Again, the MMI does not go below 3 as this is the lowest value in the FRR system. Comparing this map to the Polynomial Ridge Regression map, this map has lower values of MMI close to the epicenter and then a slower tapering off of MMI values. There are values above MMI 4 extending way further than in the ridge based model. An advantage of this model is the quicker decay all the way to MMI 3.

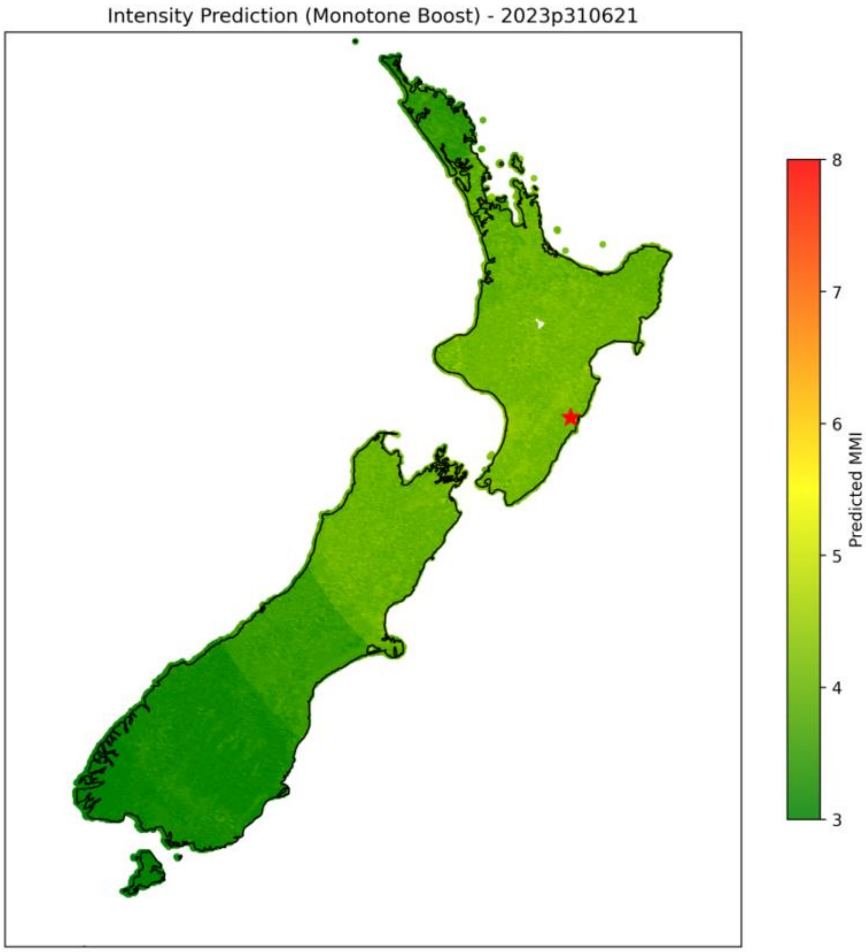

Figure 9. Predicted MMI shake map over New Zealand (Monotone Boost)

Figure 10 is the residual vs predicted MMI plot for Monotonic HistGradient BoostingRegressor. The residuals in this plot are arranged in diagonal lines the same as the degree 2 Polynomial Ridge model. The residuals seem to be generally scattered around 0 and is homoscedastic. Compared to the ridge based model, there is a wider range of residuals and prediction values here.

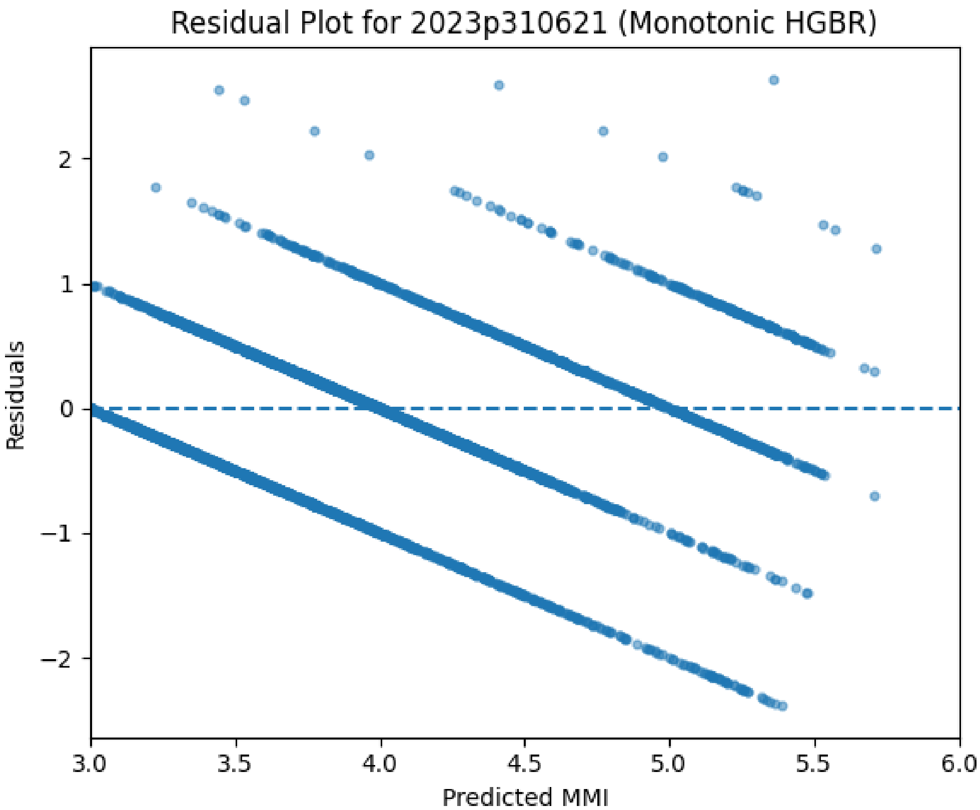

Figure 10. Residual plot for Monotone Boost

5. Discussion

The primary objective of this research project was to develop an intensity prediction model for New Zealand using FRR data, and to evaluate what modeling approaches capture the relationship between distance and shaking intensity most effectively. A secondary objective was to investigate what confounding variables influence this relationship. These results demonstrate that ridge regression with a degree 2 polynomial transform provides the most physically plausible and practical model among those tested.

5.1 Model Performance

Many of the 39 models evaluated achieved reasonable performance metrics but failed to produce intensity patterns that followed the physics of intensity attenuation. In particular the shaking intensity maps produced by these models fell into three categories, these were: predicting near average values for all areas, predicting an increase or wave pattern of intensity with distance, and generating overfit noise that contained no useful information. These models may have been able to be improved but models that were able to quickly perform well were prioritised. Producing these maps was an important part of the model selection process and highlights the limit of relying solely on performance metrics such as R^2.

After this screening and selection process there were 4 models considered acceptable, these were all regression models. Of these the degree 2 Polynomial Ridge Regression and the Monotonic HistGradientBoostingRegressor performed best. The monotonic gradient boosting model had better performance metrics, however the polynomial ridge model had a far better map which captured the physical attenuation better. The polynomial ridge model was selected as it was simpler, more interpretable and captured the physical phenomenon better.

5.2 Influence of Predictors and Equation

Appendix A has a full list of the fitted coefficients from the degree 2 polynomial ridge model. These indicate that log hypocentral distance and magnitude have the highest influence on intensity. The negative coefficient on the squared log distance term enforces rapid decay near the epicenter and slower decay further away. The negative coefficient term on magnitude is interesting but likely acts as a scaling factor offset by the log distance x magnitude coefficient. Depth primarily interacts with distance so that shallower events produce higher intensities at short distances. Vs30 has a somewhat lessened effect on the predicted outcome and seems to have the lowest effect on predicted intensity. The relatively small number of large coefficients further supports the parsimony of the Polynomial Ridge model.

Equation:

$$IMMI = -1.2544m - 1.0934d^2 + 0.8631dm + 0.5720d - 0.02286m^2 - 0.02071dh + 0.01220h + 0.01058v - 0.005886dv + 0.002329hm + 0.0002799mv + 8.98 * 10^{-5}h^2 + 9.96 * 10^{-6}hv - 3.51 * 10^{-7}v^2$$

Where:

m = moment magnitude
d = log hypocentral distance
v = Vs30
h = earthquake focal depth

Vs30 was included as a proxy for local site effects with the seismological understanding that a lower vs30 was associated with a higher intensity, however looking at a shaking map produced by this model it is clear to see that the model has learned the opposite. This behaviour is likely caused by bias inherent to the FRR dataset, as urban areas where reports are concentrated are typically located in areas with lower Vs30 values. Further research into this is needed and this could potentially signal that choosing a FRR minimum report count of 5 was still reducing the spatial coverage too much. This same problem can be seen in the Monotonic HistGradientBoostingRegressor also.

5.3 Limitations

Several limitations in the FRR dataset and modeling approach impact the results of this study, along with difficulties in researching rare natural phenomena. First, the FRR system consists of 6 cartoon images, these are supposed to align with MMI levels 3 to 8, where the full MMI scale goes from 1 to 12. As a result of this the minimum intensity level that these models can predict is MMI 3, this is inaccurate to the physical reality and a work around is difficult as there are areas which are MMI 3 so an MMI 3 cutoff can not be applied. This floor can be seen in all produced maps but does not affect any of the performance metrics.

Another associated limitation is that there is no ‘Not Felt’ option in the FRR survey. Without this inclusion far distances are forced to MMI 3, it would also help with dealing with the different reasons why areas do not have felt reports, either lack of people to produce reports or lack of any shaking.

The FRR dataset is crowdsourced from humans that are untrained in assessing earthquake intensity. As a result reported intensities are affected by individual perception, biases, prior experience with earthquakes, building characteristics, and other environmental factors. Some of these could potentially be included in a future model but would require accessing building data and coming up with a way to model how earthquake experiences affect a population's perspective toward more earthquakes. Without these inclusions in a model this variability creates extra noise and puts a limit on how good produced models can be. The FRR dataset is also concentrated around population centers; there are large areas of New Zealand with very little or no FRRs. This means that the model will be fitted to the geological conditions associated with these large population centers and predictions for sparsely populated areas will be less reliable.

A limitation inherent to modeling earthquakes is the rarity of high magnitude events, this limits the availability of balanced training data especially for high MMI reports as these are associated with high magnitude events. As a result the model performance at higher magnitudes is less certain and intensities may be more influenced by other predictors. This is a major limitation as intensity prediction equations would be most useful with high magnitude events.

Finally Vs30 is a simplification of local sight effects and there are many more related variables that could be considered for future research, these include elevation, terrain blocking, direction, fault type, and earthquake duration. Changing hypocentral distance for distance to rupture could also improve the performance.

6. Conclusions

This research project developed and evaluated a range of models for predicting shaking intensity using FRR data. It was demonstrated that both good performance metrics and spatial maps that make sense are required for selecting the best models.

A cleaning pipeline was implemented to remove invalid data and constrain some of the noise inherent in the crowd sourced FRR data. Mode was selected as the measure of central tendency as the MMI scale is ordinal data with discrete categories, and mode also captures the most commonly reported value without factoring in extreme values or skew.

After filtering out models with poor metrics or poor shaking maps the pool of 39 models was reduced to 4 regression based models. The two best were identified as Polynomial Ridge Regression and Monotonic HistGradientBoostingRegressor, out of these two Polynomial Ridge Regression was chosen as it was simpler and had a better intensity shaking map. This model is highly interpretable as it allows direct extraction of the predictor coefficients, allowing examination of the relationships between predictors and MMI.

The results also highlight fundamental limitations of using crowdsourced FRR data for intensity prediction, including reporting subjectivity, population bias, restricted intensity range, and the scarcity of high-magnitude events. These constraints place an upper bound on achievable model performance and reinforce the need for cautious interpretation of results, particularly at low and high intensity levels. This also prompts further research into what other variables could be used in predictions to account for this noise.

Overall this project demonstrates that physically interpretable regression models can provide meaningful intensity predictions. The predictors of hypocentral distance, magnitude, depth and Vs30 provide varying levels of influence on the intensity. Future improvements in the FRR system or access to data on other features such as earthquake type and rupture dynamics could further improve model performance.

7. Future work

Despite the limitations the results demonstrate that FRR data can be used to develop a physically meaningful and interpretable intensity attenuation model for New Zealand. The success of the polynomial ridge model indicates that relatively simple models can outperform more complicated methods in this area. Future work could extend this study and improve results by including additional attributes such as fault mechanisms, rupture geometry, population, duration and building types. This data was not accessible for this study, so permission would be needed to use this data in future research. Additionally, including ground motion or other scientific measurements alongside FRR data could help constrain attenuation behaviour at low intensities and reduce the impact of reporting bias; however, this was outside the scope of this study. In the future, the regional model can be added.

8. Acknowledgments

During our research process, we were grateful for the patient and detailed guidance provided by our supervisors, Hazel Fraser and Goldie Leung. StackOverflow, ChatGPT and Wikipedia were used to provide background knowledge and boiler plate code to build off of.

9. References

"1 in 4 chance of a magnitude 8 or greater earthquake in the next 50 Years" scientists uncover crucial knowledge about our largest and most active fault. East Coast LAB | Hikurangi Subduction Zone M9 | New Zealand. (n.d.). https://www.eastcoastlab.org.nz/home/article/216/

Boore, D. (1997). Equations for Estimating Horizontal Response Spectre and Peak Acceleration from Western North American Earthquakes. https://repo.nzsee.org.nz/bitstream/handle/nzsee/2602/Ingham.pdf?sequence=1&isAllowed=y

Borcherdt, R. (1970). *Effects of Local Geology on Ground Motion near San Francisco Bay*.

Changes to our felt report maps. GeoNet. (n.d.). https://www.geonet.org.nz/news/7AZhBTG25Ryz2DrQfLCcSF

Earthquake magnitude, energy release, and shaking intensity. USGS. (n.d.). https://www.usgs.gov/programs/earthquake-hazards/earthquake-magnitude-energy-release-and-shaking-intensity

Fraser, H., Goded, T., Ward, N., Andrews, J., & Hultquist, C. (2024). Use of felt rapid reports as a reliable data source in the production of earthquake intensity maps. *2024 International Conference on Machine Intelligence for GeoAnalytics and Remote Sensing (MIGARS)*, 1–3. https://doi.org/10.1109/migars61408.2024.10544828

Howarth, J. D., Cochran, U. A., Langridge, R. M., Clark, K., Fitzsimons, S. J., Berryman, K., Villamor, P., & Strong, D. T. (2018). Past large earthquakes on the alpine fault: Paleoseismological progress and future directions. *New Zealand Journal of Geology and Geophysics*, *61*(3), 309–328. https://doi.org/10.1080/00288306.2018.1464658

The severity of an earthquake. (n.d.). https://pubs.usgs.gov/gip/earthq4/severitygip.html

Kurnia, G. (2024, September 11). *Choosing the optimal data split for machine learning: 80/20 vs 70/30*. *Medium*. https://medium.com/@gunkurnia/choosing-the-optimal-data-split-for-machine-learning-80-20-vs-70-30-0fd266710236

GeoNet Home. (n.d.). https://felt.geonet.org.nz/

Dowrick, D. J., & Rhoades, D. A. (2005). Revised models for attenuation of Modified Mercalli intensity in New Zealand earthquakes. *Bulletin of the New Zealand Society for Earthquake Engineering*, *38*(4), 185–214. https://doi.org/10.5459/bnzsee.38.4.185-214

Masui, T. (2022, January 20). All you need to know about gradient boosting algorithm: Part 1 (Regression). Towards Data Science. https://towardsdatascience.com/all-you-need-to-know-about-gradient-boosting-algorithm-part-1-regression-2520a34a502/

GeeksforGeeks. (n.d.). Metrics for machine learning model. Retrieved January 1, 2026, from https://www.geeksforgeeks.org/machine-learning/metrics-for-machine-learning-model/

Murel, J. (n.d.). Downsampling. *IBM Think*. Retrieved January 2, 2026, from https://www.ibm.com/think/topics/downsampling

9. Appendices

Feature	Coefficient
log_hypo	+0.5720
Depth	0.0122
Magnitude	-1.2544
vs30	0.0106
log_hypo^2	-1.0934
log_hypo × depth	-0.0207
log_hypo × magnitude	0.8631
log_hypo × vs30	-0.0059
Depth^2	0.00009
Depth × magnitude	0.0023
Depth × vs30	0.00001
Magnitude^2	-0.0229
Magnitude × vs30	0.0003
vs30^2	0.0000

Appendix A. Estimated coefficients of the polynomial attenuation model